MANU DWIVEDI

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RESEARCH SUMMARY

Director of Engineering, AI/ML R&D at Tiami Networks, leading a team of about 20 engineers. My research asks how far a communication network can serve as a sensor: integrated sensing and communications (ISAC) that turns existing 5G base stations and broadcast transmitters into radar, physics-grounded RF digital twins that predict where sensing will work, anomaly detection that treats a network as a graph and learns only its normal behavior, and world models that fuse infrastructure and agent sensing to give robots and drones conditional autonomy. Each line has been taken from first experiment to field trial with operators, including a public trial with Vodafone.

RESEARCH INTERESTS

ISAC with 5G NR, broadcast TV, and Wi-Fi illuminators. Passive and bistatic radar. Radar foundation models. RF digital twins and propagation modeling. Self-supervised and diffusion-based anomaly detection with calibrated false-alarm control. World models and multi-agent autonomy over shared infrastructure.

RESEARCH LEADERSHIP

Tiami Networks

Director of Engineering, AI/ML R&D *Oct 2025 to Present*

- Lead about 20 engineers across machine learning, signal processing, RF, embedded systems, and robotics, and set the research agenda from first experiment to field deployment.
- Raised about **$6M** for sensing and ISAC research and productization, including the PIN-Cyber program and over $1M in follow-on funding for wideband RF analysis. Results reached approximately every major US and EU telecom operator as counter-UAS sensors.

RESEARCH PROGRAMS

PolyRAN: Radar Sensing from Existing 5G Base Stations **Tiami Networks**

2024 to Present

- Question: can a live 5G network sense its surroundings with no new radios and no change to the RAN vendor's software? PolyRAN reuses uplink reference signals the base station already schedules (SRS, PUSCH DMRS), read through a vendor-agnostic O-RAN E3 service model, and processes them with delay-Doppler correlation, CFAR detection, tracking, and a radar foundation model.
- **Public trial with Vodafone** at its Málaga R&D lab on live 5G: objects and people detected without disrupting calls, messages, or data. Further trials with **AirSpan** and tier-2 operators in Asia and the EU.

PolyEdge and AirRadar: Passive Radar on Signals of Opportunity **Tiami Networks**

2023 to Present

- Bistatic passive radar that reconstructs the 5G downlink (SSB, PBCH, CRS) or broadcast TV (DVB-T, ATSC) as its reference, cancels direct-path clutter (Wiener-Hopf, ECA), and detects targets in delay-Doppler. A radar foundation model handles detection and classification; no transmitter or spectrum license is needed.
- Small drones detected at about **1 km** (below 400 ft, up to 15 m/s, radar cross-section about 0.01 m^2); 5G-connected drones identified at 95% in clutter; light aircraft tracked to about **25 km** on broadcast TV against ADS-B truth.

RF Digital Twins for ISAC Planning **Tiami Networks**

2024 to Present

- City-scale RF digital twin from map and 3D building geometry. Vision-language agents infer building materials and clutter from street imagery, and a 3GPP TR 38.901 propagation model plans communication coverage and bistatic sensing coverage on one scene.
- Field validation of predicted signal strength against commercial phone telemetry in North America, Europe, and Asia across eight bands, standalone and non-standalone 5G. Representative n41 case: **4.22 dB** mean absolute error, 4.73 dB RMSE, 17,324 samples.

CoreDiff: Zero-Day Anomaly Detection on Network Graphs **Tiami Networks; PIN-Cyber**

2025 to Present

- Structures any network into a graph of directed interactions, each a windowed time series. A self-supervised encoder learns normal behavior, a latent diffusion model estimates its density, and low-density interactions are flagged, so unseen attack families are caught without signatures. Margin and hard-negative training sharpen the boundary; logistic calibration sets an explicit false-positives-per-hour operating point; Venn-Abers intervals give decision confidence. Evaluated with red-team attack runs on a network emulator plus denial-of-service and data-exfiltration scenarios.
- At the 1% anomalous tail: 5G SA about **94% F1** (time to detect 20 to 5 minutes); private networks about **98.9% F1** (mean time to detect 120 to 0.6 seconds) across 13 MITRE ATT&CK techniques.

CoreDiffWFM: World Models for Conditional Autonomy **Tiami Networks**

2025 to Present

- Question: can infrastructure give autonomy to an agent that has none (no map, no global position, poor sensing)? The world model jointly reconstructs the world from infrastructure sensing (ISAC, network positioning, fixed cameras) and each agent's own telemetry, fuses the sources with per-source provenance, and returns goals, holds, and keep-out zones for the agent's own navigation to execute. Agents' observations fill the belief where infrastructure coverage is poor.
- **Indoor:** a ground robot with no map or position is localized from fixed cameras and its lidar (offline runs), with the room rebuilt from a scan and a 3D Gaussian-splat scene prior. **Outdoor:** drone and ground-robot fleets over city geometry, where one agent's hazard replans every affected mission.
- An RF planner and a ray-traced physics model (Sionna RT) predict radio and world beliefs at each time step, explaining network behavior for an agent that must keep its link through a mission. Results: a predicted 1.000 m move observed as 0.980 m on a robot testbed (within three sigma); in simulation, fleet completion fell from 152 to 125 seconds, and ground robots finished 6 of 6 missions around a blockage versus 3 of 6.

ARCHES and RF Pattern-of-Life **Tiami Networks**

2025 to Present

- **ARCHES**: end-to-end counter-UAS response; a radar cue hands off to an interceptor drone guided by onboard vision (YOLOv8) and Remote-ID, with a staged PX4 safety process from simulation to flight. **Pattern-of-Life**: identifies and locates emitters (Wi-Fi, drone links, 5G) from raw RF without decoding, using burst fingerprinting and four-antenna angle of arrival.

EARLIER RESEARCH

Tiami Networks

Principal and Sr. Embedded ML Engineer; AI FPGA Intern *Sept 2023 to Oct 2025*

- First 5G passive sensing models: SSB-derived channel state into a deep classifier for five activities (IEEE RadarConf 2024), and person presence from 5G synchronization signals on a software-radio testbed (IEEE MILCOM 2024 demo).
- Uplink 5G pedestrian tracking with NVIDIA and Northeastern University, and ISAC fall detection with Deutsche Telekom (MWC 2025). Closed-loop network assurance with 3GPP NWDAF won the Deutsche Telekom T-Challenge 2024 award. Pedestrian and traffic monitoring for the City of Elk Grove at 99.6% accuracy from 250 m to 1.3 km.
- Cross-modal radar and camera models (patent pending); band-agnostic 5G drone detection at 95% in clutter; Open RAN sensing for NTIA NOFO 2; Arria10 and RFSoC edge sensing demos at MWC 2024.

Michigan State University

Research Assistant *May 2022 to Jun 2023*

- Generative-AI acceleration on low-power FPGAs with a **45% compute reduction** without pruning or quantization; sparse vision models that hold up under occlusion and blur; over 10,000 lines of C++ optimized to improve MSU supercomputer availability.

OpenPower Foundation at IBM

Researcher *Aug 2020 to Dec 2021*

- Open-ISA processor accelerators (Microwatt, A2O) for data access and matrix multiply; about 4x ALU throughput for GenAI workloads.

PUBLICATIONS AND PATENTS

- M. Dwivedi et al., "5G-Based Passive Radar Sensing for Human Activity Recognition using Deep Learning," *IEEE Radar Conference*, 2024.
- M. Dwivedi et al., "Person Presence Detection using 5G Synchronization Signals on SDR Testbed," *IEEE MILCOM* (demo), 2024.
- Two U.S. patents pending: indoor and outdoor RF activity sensing; cross-modal radar and camera sensing.

AWARDS

- Deutsche Telekom T-Challenge 2024, Most Integration Potential Award (EUR 25,000). Michigan State University Graduate Office Fellowship ($7,500).

TECHNICAL SKILLS

Sensing : Passive and bistatic radar, delay-Doppler, CFAR, clutter cancellation, angle of arrival, 5G positioning, 5G NR uplink and downlink signals

ML : PyTorch, self-supervised and diffusion models, vision-language agents, Gaussian splatting, world models, CUDA, TensorRT

Systems : O-RAN E3, OpenAirInterface, NVIDIA Aerial, Sionna RT, ROS 2, PX4, software radios, FPGA; Python, C/C++, MATLAB

EDUCATION

- **Michigan State University**, M.S. Electrical and Computer Engineering, GPA 3.8. Research: generative AI on FPGAs; 5G sensing.
- **SRM Institute of Science and Technology**, B.Tech. Electronics and Communication Engineering, GPA 3.8. Research: DSP, RF.